



温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3001 WHS01
Python Programming for Business

Introduction to Academic Research

Research Workshop I

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Learning Objectives

By the end of this session, you should be able to:

- Understand what academic research is
- Identify a researchable question from an observed phenomenon
- Formulate an interesting and meaningful research question
- Conduct a preliminary literature review
- Articulate the potential research contribution



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Warm-up: Reflecting on Your Research Idea

- You have completed Assignment 1, proposing a research question on the economic, business or societal impact of AI. We will invite 1–2 students to briefly share their ideas.
- Please address the following:
 - What AI-related **phenomenon** motivated your idea?
 - What is your **research question**?
 - Why do you think this question is **important to study**?
- As we listen, consider:
 - Is the research question **clear, interesting, meaningful, researchable**?
 - What **data or evidence** might help answer it?



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What Is Academic Research

- Academic research is a **systematic process** of investigating questions to **generate new knowledge** and contribute to the existing scholarship.
- Key Features
 - **Question-driven** — starts with a clear research question
 - **Theory-informed** — builds on and contributes to existing literature
 - **Evidence-based** — relies on scientific evidences



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Type	Purpose
News / Media	Report events or trends
Industry Reports	Provide practical insights for business
Academic Research	Explain why and how phenomena occur using theory and evidence

What Is a Research Question

- Characteristics of a good research question
 - **Clear** – easy to understand and well-defined
 - **Specific** – focus on a particular issue
 - **Researchable** – can be examined using scientific evidences
 - **Meaningful** – address an important phenomenon
- What make a research question interesting?
 - “**Deny certain assumptions of their audience**”(Davis 1971)
 - What seems obvious is actually wrong
 - What appears stable is actually changing
 - What seems harmful may have hidden benefits
 - What seems beneficial may have unintended consequences

[Davis, M.S., 1971. That's interesting! Towards a phenomenology of sociology and a sociology of phenomenology. Philosophy of the social sciences, 1\(2\), pp.309-344.](#)



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Unintended / Surprising
+
Reasonable

What Is NOT a Research Question?

Not every question qualifies as a research question. A research question must be **specific, researchable, and analytical**.

1) *Questions That Are Too Broad*

- ✗ What is the impact of AI on society?
- ? Too general and difficult to study

2) *Questions That Are Purely Descriptive*

- ✗ What is ChatGPT?
- ? This asks for a definition rather than investigation

3) *Questions That Are Yes/No Questions*

- ✗ Does AI improve productivity?
- ? Too simplistic; lacks depth and mechanisms
- ✓ How does AI assistance affect worker productivity across different skill levels?

4) *Questions That Are Normative or Opinion-Based*

- ✗ Is AI good or bad for society?
- ? Value judgment rather than empirical inquiry

Unintended / Surprising / Side Effect



Reasonable

Examples

Common Assumption	“Interesting” Research Question?	Example Study
AI increases productivity	Does generative AI improve productivity for all workers equally?	Brynjolfsson et al. 2023 , Noy and Zhang 2023
AI replaces human expertise	Does AI complement or replace human expertise?	Dell’Acqua et al., 2026
AI improves learning	Does AI assistance improve short-term performance but reduce long-term learning?	Shen and Tamkin 2026
AI improves decision-making	Does reliance on AI change how humans make decisions or judgments?	Research on algorithmic decision-making

Developing a Research Question

A research question typically emerges through a process of **refining a broad topic into a focused and researchable question.**

1. Start with a phenomenon

Identify a broad area of interest
E.g., AI in knowledge work

2. Narrow the focus

Specify the context, population, or outcome
E.g., AI tools used by coders, consultants, etc.

3. Identify a research problem or puzzle

Ask what is unclear, debated, or not yet well understood
E.g., short-term efficiency vs. long-term learning

4. Formulate a research question

- **Focused** – clearly defined scope
- **Researchable** – answerable using evidence
- **Relevant** – addresses an important issues

Sources:

- Monash University Library. [Developing Research Questions](#).
- Duke University Thompson Writing Program. [What Makes a Good Research Question?](#)

Example Progression

Phenomenon

Students increasingly use AI tools for coding.

Weak Question

Is AI useful for coding?

Better Question

How does AI assistance influence students' learning outcomes in programming tasks?

Even Better

How does reliance on AI coding assistants affect skill development among novice programmers?

Can you make it more interesting?

Literature Review

- A literature review is a **systematic examination of existing research** on **a topic** to understand what is already known and to identify gaps for new research.
- Purpose of a literature review
 - Understand existing knowledge on the topic (**scope matter**)
 - Identify theories, methods, and key findings
 - Locate gaps or unresolved questions
 - **Position your research within the scholarly conversation**
- Basic process
 - *Search relevant studies:* Google Scholar, journals/conferences in your domain, SSRN, NBER
 - *Read efficiently:* abstract, introduction, and conclusions
 - *Identify key themes:* what do researchers agree or disagree on?
 - *Find the research gap:* what question remains unanswered?

Tools for Conducting Literature Review

1. Keywords Search – Identify Relevant Papers

- Google Scholar, Web of Science, University Library databases
- Journal databases: INFORMS PubsOnline, ScienceDirect, JSTOR
- Working papers repositories: SSRN, NBER, arXiv
- Core Journals and Conferences in your field

2. Citation Exploration

- Backward search: review references cited in a paper
- Forward search: find newer papers citing that paper
- [Connected Papers](#)

3. Reference Management Tools – Organize and manage research papers

- Zotero, Mendeley, EndNote

4. Note-Taking and Organization – Record key insights from each paper

- Excel, Google Sheets, OneNote, Word

5. AI Tools – Assist with paper discovery, summarization, and literature synthesis

The most important aspect is developing
a clear framework to synthesize the
literature and position your research.

How to Synthesize the Literature

Review section

vs

Literature
review paper

A literature review should **synthesize ideas across studies**, not simply summarize individual papers.

1. *Identify Key Themes*

- Organize studies around major concepts, theories, or mechanisms (*Webster & Watson, 2002*)
- Example: AI and productivity, AI and skill development, human-AI collaboration

2. *Compare and Contrast Findings*

- Examine patterns across studies: What findings converge? Where do results diverge? What explanations are proposed? (*Booth, Colomb, & Williams, 2016*)

3. *Identify Research Gaps*

- Look for: unexplored contexts, inconsistent findings, missing mechanisms (*Creswell & Creswell, 2018*)

4. *Position Your Study*

- Explain how your research addresses a gap, tests a new mechanism, or extends existing theory

A strong literature review builds a **coherent narrative** explaining what we know and **why your study is needed**.

Arguing Contribution

- A strong research contribution explains how a study advances existing theory or knowledge.
- A theoretical contribution should clarify:
 - **What** — the key concepts or variables being studied
 - **How** — the relationships among these variables
 - **Why** — the underlying logic or mechanism explaining these relationships
 - **Who / Where / When** — the boundary conditions of the theory
- What counts as a meaningful contribution?
 - Introducing new relationships or mechanisms
 - Challenging existing assumptions or explanations
 - Extending theories to new contexts or conditions
 - Revealing limitations or boundary conditions of existing theories

[Whetten, D.A., 1989. What constitutes a theoretical contribution?. *Academy of management review*, 14\(4\), pp.490-495.](#)

Revisiting Your Research Idea

Now revisit the research question you proposed in Assignment 1.

- After today's discussion, reflect on the following:
 - *Phenomenon*: What AI-related phenomenon motivates your study?
 - *Research Question*: Is your research question clear, specific, and researchable?
 - *Literature*: What existing studies relate to your topic?
 - *Contribution*: What new insight could your study provide?
- Reflection
 - Can you refine your research question to make it:
 - more **focused**
 - more **interesting**
 - more **meaningful** to existing research?

Final Thoughts

- Academic research often begins with curiosity about a phenomenon.
- A research question is **not inherently interesting** – it becomes interesting through how it is framed and developed.
- Publication does not necessarily mean a study is interesting or valuable.
- Developing **research taste** – the ability to recognize meaningful and interesting questions – is important.
- Research taste is **partly subjective** and influenced by the interests and expectations of the scholarly community.
- Research should remain creative and intellectually enjoyable – today's frameworks are tools, not constraints.
- Continue to think critically and creatively, and do not hesitate to explore ideas beyond conventional perspectives.

WKU Writing Center

essays, speeches, presentation
slides, webpages, blogs,
podcasts, and videos

Writing Assistance Offered

- 1) Idea generation and brainstorming strategies
- 2) Research and using the library database
- 3) Reading journal articles and taking notes
- 4) Citation
- 5) Outlining and drafting
- 6) Formatting according style guides
- 7) Proofreading and using spelling/grammar checkers
- 8) Revision and collaborative writing project management

Appointment Procedure

- 1) Visit the WKU Resource Booking Page: [Resource Booker](#)
- 2) Select “WKU Writing Center”
- 3) Choose a faculty, graduate student, or peer consultant/tutor based on available time slots
- 4) Email assignment sheets or drafts to the consultant/tutor prior to the appointment
- 5) Attend the session at GEH A205
- 6) Sign out by scanning the exit QR code upon completion